

RUIKE LYU

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Research Interests: Power and energy systems; demand flexibility from large loads; electricity markets.

RESEARCH APPOINTMENTS

- **Department of Earth and Environmental Engineering, Columbia University** Oct. 2026–Present
Postdoctoral Research Scientist, Postdoctoral Advisor: Prof. Bolun Xu New York, NY
- **Dept. of Electrical and Electronic Engineering, The Hong Kong Polytechnic University** Jul. 2026–Sept. 2026
Postdoctoral Research Associate Hong Kong
- **Andlinger Center for Energy and the Environment, Princeton University** Mar. 2025–Feb. 2026
Visiting Research Collaborator, Faculty Host: Prof. Jesse Jenkins Princeton, NJ

EDUCATION

- **Department of Electrical Engineering, Tsinghua University** Sept. 2021–Jun. 2026
Ph.D. in Electrical Engineering, with the university's highest honor for graduates; PhD Advisor: Prof. Chongqing Kang Beijing, China
- **Department of Electrical Engineering, Tsinghua University** Sept. 2017–Jun. 2021
B.E. in Electrical Engineering, with honors Beijing, China
- **School of Economics and Management, Tsinghua University** Sept. 2018–Jun. 2021
Bachelor of Business Administration (Second Degree) Beijing, China

SELECTED HONORS AND AWARDS

- **Outstanding Doctoral Dissertation**, Tsinghua University 2026
- **Finalist, Top 10 Rising Stars in Research**, Tsinghua University 2026
- **Best Presentation Award (one of two)**, IEEE PES Ph.D. Dissertation Challenge 2025
- **Best Oral Presentation Award**, PCCC Student Networking Session, IEEE PES General Meeting 2025
- **National Scholarship for Graduate Students**, Ministry of Education of China 2025, 2023
- **Finalist, University-wide Top 10 Students**, Tsinghua University 2025
- **Outstanding Senior Thesis**, Tsinghua University 2021

PUBLICATIONS

J=JOURNAL ARTICLE, C=CONFERENCE PAPER

Manuscripts under Review

- [J.15] **R. Lyu** and H. Guo, "Demand flexibility must account for consumer-side capital costs," *Nature Reviews Clean Technology*, under review.
- [J.14] **R. Lyu** and T. Li, "Do We Still Need Demand Flexibility as Batteries Become Cheaper? A Levelized Cost Perspective," *Energy Conversion and Management*, under review. [arXiv:2608.07348](https://arxiv.org/abs/2608.07348).

Journal Articles

- [J.13] **R. Lyu**, A. Li, J. Wang, H. Luo, E. Du, H. Guo, C. Kang, and J. Jenkins, "Industrial overcapacity can enable seasonal flexibility in electricity use," *Nature Energy*, **11**, 1203–1215 (2026). (**Cover Article**)
- [J.12] Y. Shen, **R. Lyu**, S. Wang, J. Dai, H. Guo, and C. Kang, "Thermal-Mass Coupled Modeling and Optimal Scheduling of Aluminum Smelting Loads for Safe Demand Response Participation," *IEEE Transactions on Smart Grid*, in press.
- [J.11] X. Su, **R. Lyu**, Y. Bai, X. Wang, W. Zhao, and H. Guo, "A Method for Modeling the Feasible Region of Industrial Production Processes Based on the Optimal Adjustable Load Model," *Automation of Electric Power Systems*, 2026, 50 (02): 18-29.
- [J.10] **R. Lyu**, H. Guo, G. Strbac and C. Kang, "Data-Driven Dimension Reduction for Industrial Load Modeling Using Inverse Optimization," *IEEE Transactions on Smart Grid*, vol. 16, no. 3 (2025): 2695-2698.
- [J.9] **R. Lyu**, H. Guo, Q. Tang, Q. Chen, and C. Kang, "Production Scheduling Identification: An Inverse Optimization Approach for Industrial Load Modeling Using Smart Meter Data," *IEEE Transactions on Smart Grid*, vol. 16, no. 2 (2025): 1207-1220.
- [J.8] **R. Lyu**, X. Su, E. Du, H. Guo, Q. Chen and C. Kang, "Efficient Scheduling of Discrete Industrial Processes through Continuous Modeling," *IEEE Transactions on Smart Grid*, vol. 16, no. 6 (2025): 4726-4740.

- [J.7] Q. Chen, **R. Lyu**, H. Guo, and X. Su, "Real-Time Operation Strategy of Virtual Power Plants With Optimal Power Disaggregation Among Heterogeneous Resources," *Applied Energy*, vol. 361 (2024): 122876.
- [J.6] X. Su, **R. Lyu**, H. Guo, and Q. Chen, "A Method for Optimal Selection of High-Capacity Industrial Users for Demand Response Based on Load Step Data Processing Mode," *Electric Power*, 2024, 57(01): 18-29.
- [J.5] **R. Lyu**, H. Guo, K. Zheng, M. Sun, and Q. Chen, "Co-Optimizing Bidding and Power Allocation of an EV Aggregator Providing Real-Time Frequency Regulation Service," *IEEE Transactions on Smart Grid*, vol. 14, no. 6 (2023): 4594-4606.
- [J.4] Q. Chen, **R. Lyu**, H. Guo, H. Jia, Y. Ding, Y. Wang, and C. Kang, "Electricity User Behavior Modeling for Demand Response: Research Status Quo and Applications," *Electric Power Automation Equipment*, 2023, 43 (10): 23-37.
- [J.3] **R. Lyu**, Y. Gu, and Q. Chen, "Electric Vehicle Charging Right Trading: Concept, Mechanism, and Methodology," *IEEE Transactions on Smart Grid*, vol. 13, no. 4 (2022): 3094-3105.
- [J.2] Q. Chen, **R. Lyu**, Q. Tang, K. Li, H. Gao, and H. Guo, "Emergency Response of Electricity Market from Perspectives of Suspension of Spot Market in Australia," *Automation of Electric Power Systems*, 2022, 46 (16): 214-223.
- [J.1] Q. Chen, X. Fang, H. Guo, K. Zheng, Q. Tang, **R. Lyu**, K. Pan, P. Palensky, D. S. Kirschen, and C. Kang, "The Competition and Equilibrium in Power Markets Under Decarbonization And Decentralization," *iEnergy*, vol. 1, no. 2 (2022): 188-203.

Conference Papers

- [C.10] J. Li, **R. Lyu**, Y. Zhang, X. Cha, K. Zheng, and H. Guo, "Value-Based Industrial Load Shedding with Supply Chain Coordination Reduces Electricity Cost," *2026 IEEE PES International Meeting*, Hong Kong, 2026, pp. 1-5.
- [C.9] J. Zhang, **R. Lyu**, X. You, J. Wang, Y. Cai, and H. Guo, "When Will Real-Time Pricing Outperform Time-of-Use Pricing Significantly?," *2026 IEEE PES International Meeting*, Hong Kong, 2026, pp. 1-5.
- [C.8] H. Huang, **R. Lyu**, C. Feng, H. Zhong, H. B. Gooi, B. Li and R. Liang, "Learning for Feasible Region on Coal Mine Virtual Power Plants with Imperfect Information," *2025 IEEE PES General Meeting*, Austin, TX, USA, 2025, pp. 1-5.
- [C.7] Q. Liu, **R. Lyu**, Z. Zhai, Y. Shen, X. Liu and H. Guo, "Integrating Fast-response Capability into Virtual Power Plant Operation for Ancillary Services," *2025 IEEE Powertech*, Kiel, Germany.
- [C.6] R. Chen, Z. Tang, **R. Lyu**, Q. Zheng, H. Song and H. Guo, "Combining AI and Simulation to Assess Building Demand Response Potential at Scale," *2025 5th International Conference on Advances in Electrical, Electronics and Computing Technology (EECT)*, Guangzhou, China, 2025, pp. 1-6. **(Best Presentation)**
- [C.5] Y. Shen, **R. Lyu**, H. Guo and C. Kang, "An Improved Modeling Method for Electrolyte Aluminum Loads Considering Thermal Balance and Flexible Regulation Cost," *2025 IEEE International Conference on Power Systems and Smart Grid Technologies (PSSGT)*, Chongqing, China, 2025, pp. 331-337. **(Best Presentation)**
- [C.4] **R. Lyu**, H. Guo and Q. Chen, "Approximating Energy-Regulation Feasible Regions of Virtual Power Plants: A Data-Driven Inverse Optimization Approach," *2024 IEEE Power & Energy Society General Meeting (PESGM)*, Seattle, WA, USA, 2024, pp. 1-5.
- [C.3] A. Luo, **R. Lyu**, H. Guo, Y. Cai and Q. Chen, "An Incentive-Compatible VPP Profit Allocation Model Considering the Operating Characteristics of Air Conditioning Loads," *2024 7th International Conference on Energy, Electrical and Power Engineering (CEEPE)*, Yangzhou, China, 2024, pp. 1270-1278. **(Best Presentation)**
- [C.2] L. Su, **R. Lyu**, H. Guo, Y. Cai and Q. Chen, "Coordinating Air Conditioning Load Clusters to Provide Frequency Regulation Using Setpoint Changes," *2024 7th International Conference on Energy, Electrical and Power Engineering (CEEPE)*, Yangzhou, China, 2024, pp. 905-910.
- [C.1] **R. Lyu**, H. Guo, Y. Zheng, Y. Bai and Q. Chen, "LSTN: A Linear Model of Industrial Production Process for Demand Response," *2023 IEEE PES Innovative Smart Grid Technologies Europe (ISGT EUROPE)*, Grenoble, France, 2023, pp. 1-5.

SELECTED RESEARCH PROJECTS

- **Net Zero China** 2024–Present
Core Team Member; joint initiative of Tsinghua University and Princeton University
 - Develop sector- and region-specific industrial energy-demand and decarbonization scenarios using EnergyPlus.
 - Implement aluminum, ammonia, and methanol production models in the open-source MacroEnergy.jl energy-system planning framework.
- **Energy Conservation and Supply-Demand Interaction Technology for High-Energy-Consumption Industrial Users**
Core Team Member; National Key Research and Development Program of China, Grant No. 2023YFB2407300, 2023–2026
- **Key Technologies of Interactive Regulation of Large-Scale Flexible Resources Aggregation in Virtual Power Plants**
Core Team Member; National Key Research and Development Program of China, Grant No. 2021YFB2401200, 2021–2025

INDUSTRY COLLABORATION

- **Industrial Energy Management for a BYD Automotive Manufacturing Plant** 2024
Implementation Lead; State Power Investment Corporation (SPIC) Chongqing, China
 - Developed and implemented an optimization strategy coordinating a 60 MW manufacturing load, a 60 MW/240 MWh battery energy storage system, and 10 MW of solar generation.
 - Projected annual operating-cost savings of approximately US\$4 million.

PATENT APPLICATIONS

- [PAT.2] Q. Chen, Y. Chen, K. Zheng, and **R. Lyu**, "Method for updating state of charge based on power characteristic of electrochemical model of lithium-ion battery," U.S. Patent Application No. 18/919,638 (2025).
- [PAT.1] Q. Chen, Y. Chen, K. Zheng, and **R. Lyu**, "Method for estimating state of power based on electrochemical model of lithium-ion battery," U.S. Patent Application No. 18/919,593 (2025).

INVITED SEMINARS AND TALKS

- **The Chinese University of Hong Kong** Sept. 17, 2026
Host: Prof. Yue Chen Hong Kong
- **The University of Hong Kong** Apr. 1, 2026
Host: Prof. Yi Wang Hong Kong
- **Columbia University** Feb. 9, 2026
Host: Prof. Bolun Xu New York, NY
- **Princeton University** Feb. 6, 2026
Host: Prof. Denise Mauzerall Princeton, NJ
- **Harvard University** Feb. 4, 2026
Host: Dr. Chris Nielsen Cambridge, MA
- **Massachusetts Institute of Technology (MIT)** Feb. 3, 2026
Host: Dr. Ruairidh MacDonald Cambridge, MA
- **Dartmouth College** Feb. 2, 2026
Host: Prof. Cong Chen Hanover, NH
- **New York University** Jan. 28, 2026
Host: Prof. Yuzhang Lin New York, NY
- **Stanford University** Dec. 5, 2025
Host: Prof. Adam Brandt Stanford, CA
- **University of Washington** Oct. 17, 2025
Host: Prof. Baosen Zhang Seattle, WA
- **California Institute of Technology (Caltech)** Oct. 16, 2025
Host: Prof. Steven Low Pasadena, CA
- **University of California, Riverside** Sept. 26, 2025
Host: Prof. Nanpeng Yu Riverside, CA

TEACHING AND MENTORING

- **Department of Electrical Engineering, Tsinghua University** 2023–2024
Teaching Assistant for Energy Systems Operation, Dispatch, and Planning Beijing, China
 - Prepared course materials on energy storage, virtual power plants, and electricity markets; designed and graded all assignments.
 - Developed two open-ended optimization projects based on virtual-power-plant research, in which students formulated models and proposed their own solution methods.
- **Department of Electrical Engineering, Tsinghua University**
Graduate Research Mentoring Beijing, China
 - Co-mentored three master's students and one PhD student.

ACADEMIC AND PROFESSIONAL SERVICE

- **Peer review:** Completed 40 reviews for 10 journals, including *IEEE Transactions on Smart Grid*, *Applied Energy*, *IEEE Transactions on Power Systems*, *IEEE Internet of Things Journal*, and *Scientific Reports*.
- **Professional service:** Member of the IEEE Task Force on Systems Modelling for Electricity Market Design and the IEEE Task Force on Economics of Flexibility Models & Data; co-hosted a task-force meeting at the 2026 IEEE Power & Energy Society General Meeting and a 2026 symposium in Montreal on energy and power education, research innovation, and career development in the AI era.
- **University service:** Served as an Undergraduate Counselor at Tsinghua University; conducted more than 80 one-on-one advising sessions, organized seven faculty discussions on research and careers, and coordinated industrial site visits. Recognized as an Outstanding Undergraduate Counselor.
- **Academic community engagement:** Served as a panelist at IEEE PES Day events and led presentations and student discussions on emerging energy policies and technologies through the Princeton University Energy Association.